# MAT 216
## Week 4

1. The Number of solutions of a system of equations.

2. Whether or when such a system has a solution or how many.

3. Properties of inverse matrices

4. Number of solutions of a system of equations with parametric co-efficient

5. How to solve a system of linear equations with Gauss Jordan Elimination.

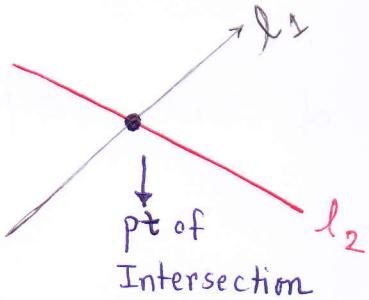
* Read the pdf Lecture (Week 4) for worked example

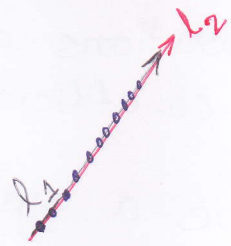
* Watch Example Tutorial of Week 4.

$\boxed{1}$ & $\boxed{2}$

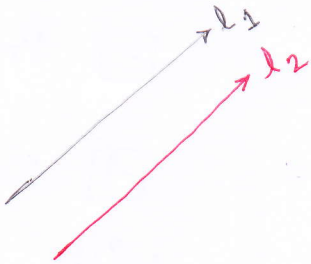
A System of Linear Eqn

- Consistent
  - Unique/One Solution
  - Many Solution
- Inconsistent
  - No Solution




infinitely many points of intersection




parallel lines never intersect with each other

$\boxed{\text{No Solution}}$

$$x - 3y + z = 4$$
$$-x + 2y - 5z = 3$$
$$5x - 13y + 13z = 8$$

$$\left[\begin{array}{ccc|c} 1 & -3 & 1 & 4 \\ -1 & 2 & -5 & 3 \\ 5 & -13 & 13 & 8 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & -3 & 1 & 4 \\ 0 & -1 & -4 & 7 \\ 0 & 2 & 8 & -12 \end{array}\right] \begin{array}{l} \\ R_2 + R_1 \\ R_3 - 5R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & -3 & 1 & 4 \\ 0 & 1 & -4 & 7 \\ 0 & 0 & 0 & -6 \end{array}\right] R_3 + 2R_2$$

$R_3$: $0 \cdot z = -6 \rightarrow 0 = -6$ False

$\therefore$ The system is inconsistent

# Many Solution

$$x+y+z=7$$
$$3x-2y-z=4$$
$$x+6y+5z=24$$

$$\longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 3 & -2 & -1 & 4 \\ 1 & 6 & 5 & 24 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 0 & -5 & -4 & -17 \\ 0 & 5 & 4 & 17 \end{array}\right] \begin{array}{l} \\ R_2-3R_1 \\ R_3-R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 0 & -5 & -4 & -17 \\ 0 & 0 & 0 & 0 \end{array}\right] R_3+R_2$$

$R_3$: $0 \cdot z = 0 \rightarrow 0=0$ True

$z$ is a free variable

Let $z=p$ any arbitrary value

$R_2$: $-5y-4z=-17$

$5y+4(p)=17$

$5y=17-4p$

$y=\dfrac{17-4p}{5}$

$R_1$: $x+y+z=7$

$x+\dfrac{17-4p}{5}+p=7$

$x=7-\dfrac{17-4p}{5}-p$

$=7-\dfrac{17}{5}+\dfrac{4p}{5}-p$

$x=\dfrac{18}{5}-\dfrac{p}{5}=\dfrac{18-p}{5}$

$\therefore (x,y,z)=\left(\dfrac{18-p}{5}, \dfrac{17-4p}{5}, p\right)$ while $p \in \mathbb{R}$.

The system has many solution & it is consistent.

## Unique Solution

$$x+y+z=2$$
$$6x-4y+5z=31$$
$$5x+2y+2z=13$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 6 & -4 & 5 & 31 \\ 5 & 2 & 2 & 13 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -10 & -1 & 19 \\ 0 & -3 & -3 & 3 \end{array}\right] \begin{array}{l} \\ R_2-6R_1 \\ R_3-5R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 10 & 1 & -19 \\ 0 & 1 & 1 & -1 \end{array}\right] \begin{array}{l} \\ R_2 \div (-1) \\ R_3 \div (-3) \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 10 & 1 & -19 \\ 0 & 0 & 9 & 9 \end{array}\right] \begin{array}{l} \\ \\ 10R_3-R_2 \end{array}$$

$R_3$: $9z=9 \rightarrow z=1$

$R_2$: $10y+z=-19 \rightarrow 10y+(1)=-19 \rightarrow 10y=-20 \rightarrow y=-2$

$R_1$: $x+y+z=2 \rightarrow x+(-2)+(1)=2 \rightarrow x=2+2-1=3$

$(x,y,z)=(3,-2,1)$ Unique solution.

The system is consistent

# Properties of Inverse matrices:

a – Left inverse is also right inverse for a square matrix

b – Left inverse and Right inverse are the same for a square matrix: $AA^{-1} = A^{-1}A = I$

c – Product inverse formula

d – Inverse of a 2x2 matrix: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{\text{Inverse}} \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

e – Inverse of a diagonal matrix

## a Left inverse is also right inverse for a square matrix:

$$Ax = b$$

$$\underbrace{A^{-1}A}_{I}x = A^{-1}b \quad [A^{-1} \text{ is the left inverse of } A]$$

$$x = A^{-1}b$$

$$Ax = AA^{-1}b \quad [\text{multiply both sides by } A]$$

$$b = \underbrace{AA^{-1}}_{\text{It must be } I}b \quad [\because Ax = b]$$

$\therefore b = b$

$$\underbrace{A^{-1}A}_{\text{Left inverse}} = \underbrace{AA^{-1}}_{\text{Right inverse}} = I$$

# Inverse of Rectangular Matrix

Only square matrix (if it is invertible) can have inverse.

$$AB = BA = I \ , \quad B = A^{-1}$$

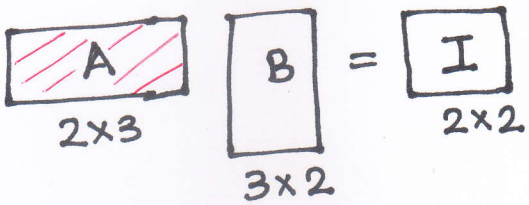
For Rectangular Matrix: $A_{m \times n}$

**Case 1:** $m < n$ Inverse of wide Matrix

Then $B_{n \times m}$ will be the right inverse of $A_{m \times n}$

Such that $AB = I_m$

example: let $m = 2$, $n = 3$

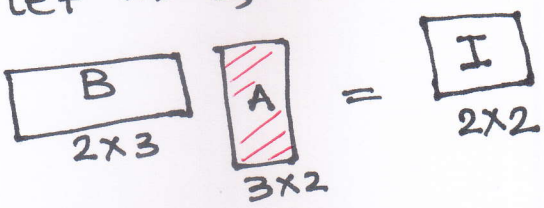



**Case 2:** $m > n$ Inverse of tall Matrix

Then $B_{n \times m}$ will be the left inverse of $A_{m \times n}$

such that $BA = I_n$

example: let $m = 3$, $n = 2$




NOTE: These two methods are partial Inverse

Example from <u>Video Lecture 8</u>

A tall matrix cannot have right inverse.
It can have left inverse in some cases.

Consider $A = \begin{bmatrix} 2 & -1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ $\quad B = A^{-1} = ?$

A is a $\boxed{3 \times 2}$ matrix

$\therefore$ I should be $I_3$

$$AB = I$$

$$\begin{bmatrix} 2 & -1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$3 \times 2 \qquad 2 \times 3 \qquad 3 \times 3$

$$\begin{bmatrix} 2 & -1 \\ 0 & 1 \\ \boxed{0 \quad 0} \end{bmatrix} \begin{bmatrix} a & b & \boxed{c} \\ d & e & \boxed{f} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \boxed{1} \end{bmatrix}$$

But $0 \cdot c + 0 \cdot f \neq 1$

Hence the tall matrix A does not have the <u>right inverse</u> B.

For a square matrix, the inverse that works on the left side of the matrix also works on the right side of the matrix

$$\underset{\substack{\downarrow \\ \text{left} \\ \text{inverse}}}{BA} = \underset{\substack{\downarrow \\ \text{right} \\ \text{inverse}}}{AB} = I; \quad B = A^{-1}$$

Example of a non square matrix:

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \qquad A^{-1} = ?$$

Let's figure out the size of $I$ and then $A^{-1}$.

$$A A^{-1} = I$$

$$\underset{2\times 3}{\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}} \underset{3\times 2}{\begin{bmatrix} ? \end{bmatrix}} = \underset{2\times 2}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}$$

Let's fill up $A^{-1}$ with some mental math

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$a - c = 1$$
$$c + e = 0$$

Add

$$a + e = 1$$

either $a = 1$, $e = 0$ or $e = 1$, $a = 0$

$$b - d = 0$$
$$d + f = 1$$
$$b + f = 1$$

either $b = 1$, $f = 0$ or $b = 0$, $f = 1$

I will consider $c = 0$, $d = 0$

Let $A^{-1} = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$

Check

$$\underbrace{\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}}_{\boxed{A}} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}}_{\boxed{A^{-1}}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\boxed{I_2}}$$

✓✓

It worked

Does it work on Left?

<u>Try</u>

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}}_{\boxed{A^{-1}}} \underbrace{\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}}_{\boxed{A}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$\downarrow$

Did not work

$$A^{-1} A \neq I$$

## [c] Product Inverse Formula:

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$$

Consider:

$$\begin{aligned}(ABC)(ABC)^{-1} &= (ABC)C^{-1}B^{-1}A^{-1} \\ &= AB(\underbrace{CC^{-1}}_{I})B^{-1}A^{-1} \\ &= A(\underbrace{BB^{-1}}_{I})A^{-1} \\ &= AA^{-1} \\ &= I\end{aligned}$$

## [e] Inverse of a diagonal Matrix:

$$\begin{bmatrix} d_1 & \cdots & & 0 \\ 0 & d_2 & \cdots & \vdots \\ 0 & \cdots & d_3 & 0 \\ 0 & \cdots & 0 & d_n \end{bmatrix} \longrightarrow \begin{bmatrix} \frac{1}{d_1} & \cdots & & 0 \\ 0 & 1/d_2 & \cdots & 0 \\ \vdots & \cdots & 1/d_3 & 0 \\ 0 & \cdots & & 1/d_n \end{bmatrix}$$

Diagonal matrix

Inverse of a diagonal matrix

$d_i \neq 0$ for all $i \in [1, n]$

A square matrix needs to have all non-zero pivots to be invertible.

$\boxed{4}$

For what values of $\lambda$ and $\mu$ the following system of linear eqn has (i) no solution (ii) many solution (iii) unique solution

$$x+y+z=6$$
$$x+2y+3z=10$$
$$x+2y+\lambda z=\mu$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array}\right] \begin{array}{l} \\ R_2-R_1 \\ R_3-R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 6 \\ 0 & \boxed{1} & 2 & 4 \\ 0 & 0 & \boxed{\lambda-3} & \mu-10 \end{array}\right]$$

Considering $R_3: (\lambda-3)z = \mu-10$

- $\lambda$ & $\mu$ are parameter
- $(\lambda-3)$ is the coefficient of $z$
- $\lambda$ & $\mu$ are not variables

$\boxed{\text{No solution}}$

if $\lambda=3$ & $\mu \neq 10$

then $R_3: (\lambda-3)z = \underbrace{\mu-10}_{\text{Non zero}}$

$$0 \cdot z = \mu-10$$
$$0 = \mu-10 \longrightarrow \text{False}$$

Hence for No solution $\lambda=3$, $\mu \neq 10$

$\boxed{\text{Many Solution}}$

if $\lambda=3$ & $\mu=10$ then $R_3: (\lambda-3)z = \mu-10$

$$0 \cdot z = 0$$
$$0 = 0$$
True

Now we have "z" as free variable. $C_3$ is dependent.

$\therefore$ Many solution while $\lambda=3$, $\mu=10$

$\boxed{\text{Unique Solution}}$

if $\lambda \neq 3$ & $\mu \in \mathbb{R}$ then $R_3: (\lambda-3)z = \mu-10$

or $z = \dfrac{\mu-10}{\lambda-3}$

So we have unique solution for "z"

Now by back substitution we can achieve unique solution for "x" & "y" $\because$ we have pivots $1, 1, \lambda-3$.

$\boxed{5}$

Solve the following system of linear eqn using Gauss Jordan Method:

$$-2x_3 + 7x_5 = 12$$

$$2x_1 + 4x_2 - 10x_3 + 6x_4 + 12x_5 = 28$$

$$2x_1 + 4x_2 - 5x_3 + 6x_4 - 5x_5 = -1$$

$$\begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \end{array} \\ \left[\begin{array}{ccccc|c} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{array}\right] \end{array}$$

$$= \left[\begin{array}{ccccc|c} 2 & 4 & -10 & 6 & 12 & 28 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{array}\right] \quad R_1, R_2 \text{ interchange}$$

$$= \left[\begin{array}{ccccc|c} \boxed{1} & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{array}\right] \quad R_1 \div 2$$

$$= \left[\begin{array}{ccccc|c} \boxed{1} & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & \boxed{-2} & 0 & 7 & 12 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{array}\right] \quad R_3 - 2R_1$$

$$= \left[\begin{array}{ccccc|c} \boxed{2} & 4 & 0 & 6 & -23 & -32 \\ 0 & 0 & \boxed{-2} & 0 & 7 & 12 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{array}\right] \quad \begin{array}{l} 2R_1 - 5R_2 \\ \\ 2R_3 + 5R_2 \end{array}$$

$$= \left[\begin{array}{ccccc|c} \boxed{2} & 2 & 0 & 6 & 0 & 14 \\ 0 & 0 & -2 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 & \boxed{1} & 2 \end{array}\right] \begin{array}{l} R_1 + 23R_3 \\ R_2 - 7R_3 \\ \phantom{0} \end{array}$$

$$= \left[\begin{array}{ccccc|c} \boxed{1} & 1 & 0 & 3 & 0 & 7 \\ 0 & 0 & \boxed{1} & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & \boxed{1} & 2 \end{array}\right] \begin{array}{l} R_1 \div 2 \\ R_2 \div (-2) \\ \phantom{0} \end{array}$$

$R_3$: $x_5 = 2$

$R_2$: $x_3 = 1$

$R_1$: $x_1 + x_2 + 3x_4 = 7$

$x_1 + a + 3b = 7$

$x_1 = 7 - a - 3b$

$x_2$ & $x_4$ are free variables

let $x_2 = a$

$x_4 = b$

any arbitrary value

$\therefore (x_1, x_2, x_3, x_4, x_5) = (7-a-3b, a, 1, b, 2)$

while $a \in \mathbb{R}$, $b \in \mathbb{R}$.